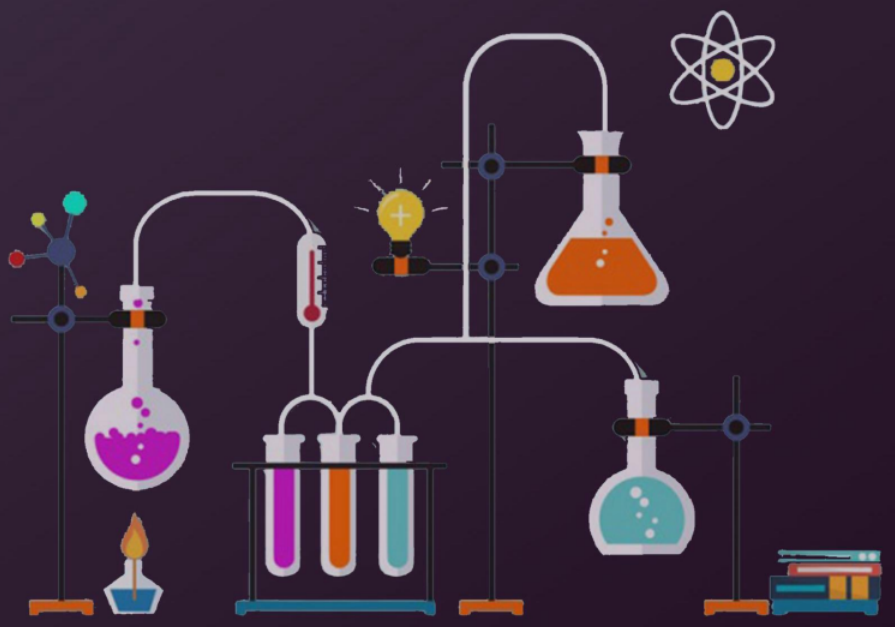
CHEMISTRY

Hydrogen



SHEET

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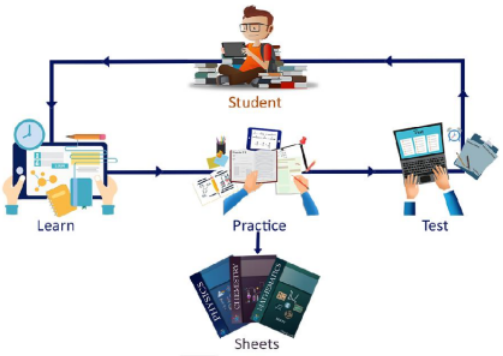
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Hydrogen	1	1	1-3	
2	Dihydrogen and it's properties	2-4			1
3	Hydrides	5	7-9		
4	Water: The necessity of life	6-7	2-4	4	
5	Water hardness and it's removal	8-11		5-11	6,9
6	Hydrogen Peroxide and Heavy Water	12-23	5,10,15	12-19	2,3-5
7	ACC		6,11-14	20-23	7,8

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Hydrogen

Q1. The nuclei of tritium atom would contain neutrons :-

(A) 1 (B) 2

(C) 3 (D) 6

Dihydrogen and It's Properties

Q2. Water gas is a mixture of :

(A) CO and H_2O

(B) CO and N_2

(C) CO and H_2

(D) CO and CH_4

Q3. Hydrogen resembles halogens in many respects for which several factors are responsible. Of the following factors which one is most important in this respect ?

(A) Its tendency to lose an electron to form a cation.

(B) Its tendency to gain a single electron in its valence shell to attain stable electronic configuration.

(C) Its low negative electron gain enthalpy value.

(D) Its small size.

Q4. Elements of which of the following group(s) of periodic table do not form hydrides.

(A) Groups 7, 8, 9

(B) Group 13

(C) Groups 15, 16, 17

(D) Group 14

Hydrides

Q5. Metal hydrides are ionic, covalent or molecular in nature. Among LiH, NaH, KH, RbH, CsH, the correct order of increasing ionic character is

(A) LiH > NaH > CsH > KH > RbH

(B) LiH < NaH < KH < RbH < CsH

(C) RbH > CsH > NaH > KH > LiH

(D) NaH > CsH > RbH > LiH > KH

Water : The Necessity of life

Q6. Both temporary and permanent hardness in water is removed by :-

(A) Boiling (B) Filtration\

(C) Distillation (D) Decantation

Q7. High boiling point of water is due to :-

(A) Its high specific heat

(B) Hydrogen bonding

(C) High dielectric constant

(D) Low dissociation constant

Water Hardness and It's Removal

Q8. Which of the following ions will cause hardness in water sample?

(A) Ca^{2+} (B) Na^{+}

(C) Cl^{-} (D) K^{+}

Q9. When zeolite is treated with hard water the Na^{+} Ions are exchanged with :-

(A) H^{+} Ions

(B) Ca^{+2} Ions

(C) Mg^{+2} Ions

(D) Both Ca^{+2} & Mg^{+2}

Q10. Permanent hardness can not removed by :-

(A) Calgon method

(B) Zeolite

(C) Clark method

(D) Ion exchange resins

Q11. Water softening by Clarke's process uses
(A) Calcium bicarbonate
(B) Sodium bicarbonate
(C) Potash Alum
(D) Calcium hydroxide

Hydrogen Peroxide and Heavy Water

Q12. Hydrogen peroxide is _____.
(A) only an oxidising agent
(B) only a reducing agent
(C) both an oxidising and a reducing agent
(D) neither oxidising nor reducing agent

Q13. When sodium peroxide is treated with dilute sulphuric acid, we get _____.
(A) sodium sulphate and water
(B) sodium sulphate and oxygen
(C) sodium sulphate, hydrogen and oxygen
(D) sodium sulphate and hydrogen peroxide

Q14. Water obtained by purification with organic ion exchange resins is-
(A) Pure water
(B) Free from only Ca^{2+}, Mg^{2+} ions
(C) Free from HCO_3^-, SO_4^{2-} and Cl^- ions only
(D) None of these

Q15. Which of the following compound affects mercury
(A) D_2O (B) H_2O
(C) O_3 (D) dil. HCl

Q16. Pure hydrogen peroxide is an unstable liquid and decomposes into water and oxygen on long standing or heating. Its decomposition is an example of :–
(A) auto-oxidaiton
(B) auto-reduction
(C) disproportionation
(D) autoprotolysis

Q17. Hydrogen peroxide reacts with ice-cold acidified potassium dichromate solution in ethene to form _____ which dissolves in ether producing a blue coloration.
(A) CrO_3 (B) CrO_5
(C) Cr_2O_3 (D) CrO_2

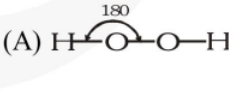
Q18. H_2O_2 is manufactured these day By :-
(A) Action of H_2SO_4 on BaO_2
(B) Action of H_2SO_4 on Na_2O_2
(C) By electrolysis of 50% H_2SO_4
(D) By Burning H_2, in excess of O_2

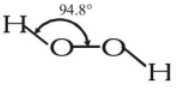
Q19. Oil painting are washed away by :-
(A) H_2O_2 (B) H_2O
(C) D_2O (D) Na_2O

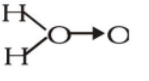
Q20. Heavy water is obtained by :-
(A) Boiling water
(B) Fractional distillation of H_2O
(C) Prolonged electrolysis of H_2O
(D) Heating H_2O_2

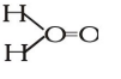
Q21. Molecular weight of heavy water is :- (in amu)
(A) 19 (B) 20
(C) 17 (D) 18

Q22. Which of the following is a true structure of H_2O_2

(A) 

(B) 

(C) 

(D) 

Q23. The hybridisation of the orbitals of oxygen in H_2O_2 is :-
(A) sp^3d (B) sp
(C) sp^2 (D) sp^3

Exercise - 1A
Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following reactions increases production of dihydrogen from synthesis gas?

(A) $CH_4(g) + H_2O(g) \xrightarrow[Ni]{1270K} CO(g) + 3H_2(g)$

(B) $C(s) + H_2O(g) \xrightarrow{1270K} CO(g) + H_2(g)$

(C) $CO(g) + H_2O(g) \xrightarrow[Catalyst]{673K} CO_2(g) + H_2(g)$

(D) $C_2H_6 + 2H_2O \xrightarrow[Ni]{1270K} 2CO + 5H_2$

Q2. Which of the following statements are not true for hydrogen?

(A) It exists as diatomic molecule.

(B) It has one electron in the outermost shell.

(C) It can lose an electron to form a cation which can freely exist

(D) It forms a large number of ionic compounds by losing an electron.

Q3. Which of the following statements is correct?

(A) Hydrides of group 13 act as Lewis acids.

(B) Hydrides of group 14 are electron deficient hydrides.

(C) Hydrides of group 14 act as Lewis acids.

(D) Hydrides of group 15 act as Lewis bases.

Q4. Hardness of water may be temporary or permanent. Permanent hardness is due to the presence of

(A) Chlorides of Ca and Mg in water

(B) Sulphates of Ca and Mg in water

(C) Hydrogen carbonates of Ca and Mg in water

(D) Carbonates of alkali metals in water

Q5. Which of the following statements is/are correct.

(A) Ionic compounds are less soluble in D_2O than in H_2O.

(B) The temperature of maximum density of H_2O is less than that of D_2O.

(C) Deuterium compounds can be prepared by exchange reactions where deuterium is exchanged for hydrogen under suitable conditions.

(D) None is correct.

Q6. Some of the properties of water are described below. Which of them is/are not correct?

(A) Water is known to be a universal solvent.

(B) Hydrogen bonding is present to a large extent in liquid water.

(C) There is no hydrogen bonding in the frozen state of water.

(D) Frozen water is heavier than liquid water.

PARAGRAPH TYPE

This section contains ***PARAGRAPHS****. Based on each paragraph, there are questions. Each question has FOUR options.* ***ONLY ONE*** *of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph For Q. 7 to 9

Hydrogen combines with a number of metals and non-metals to form compounds known as hydrides.

Q7. Ionic or salt like hydrides are formed by metals with :-

(A) Low electronegativity

(B) High electronegativity

(C) Electronegative with respect to hydrogen

(D) None of these

Q8. Molecular or covalent hydrides are :-
(A) Soft
(B) Hard
(C) High electrical conductor
(D) None of these

Q9. Metallic hydrides are :-
(A) Soft
(B) Hard
(C) Stoichiometric hydrides
(D) None of these

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q10. Match the items in Column I with the relevant item in Column II.

Column I	**Column II**
(A) Hydrogen peroxide is used as a	(p) Zeolite
(B) Used in Calgon method	(q) Perhydrol
(C) Permanent hardness of water is removed by	(r) Hexametaphosphate
	(s) Propellant

Q11. Match Column I with Column II for the given properties/applications mentioned therein.

Column I	**Column II**
(A) H	(p) Used in the name of perhydrol.
(B) H_2	(q) Can be reduced to dihydrogen by NaH.
(C) H_2O	(r) Can be used in hydroformylation of olefin.
(D) H_2O_2	(s) Can be used in cutting and welding.

Q12. Correlate the items listed in Column I with those listed in Column II. Find out as many correlations as you can.

Column I	**Column II**
(A) Synthesis gas	(p) $Na_2[Na_4(PO_3)_6]$
(B) Dihydrogen	(q) Oxidising agent
(C) Heavy water	(r) Used in softening of water
(D) Calgon	(s) Reducing agent
(E) Hydrogen peroxide	(t) Stoichiometric compounds of s-block elements
(F) Salt like hydrides	(u) Produced by prolonged electrolysis of water
	(v) Zn + NaOH
	(w) Zn + dil. H_2SO_4
	(x) Synthesis of methanol
	(y) Mixture of CO and H_2

Q13. Match the column correctly –

Column I	**Column II**
(A) Two nuclei have different spin	(p) Nascent hydrogen
(B) Two nuclei have same spin	(q) Atomic hydrogen
(C) Hydrogen produced in contact with the substance to be reduced	(r) Ortho hydrogen
(D) Hydrogen obtained by subjecting molecular hydrogen to silent electric discharge at low pressure	(s) Para hydrogen

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q14. What mass of hydrogen peroxide will be present in 2 litres of a 5 molar solution? Calculate the mass of oxygen which will be liberated by the decomposition of 200 mL of this solution.

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement -1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q15 **Statement-1:** D_2O can be used for drinking purposes.

Statement-2: D_2O is toxic to living organisms.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The sum of proton & electrons in the deuterium is :-

(A) 2 (B) 4

(C) 6 (D) 3

Q2. Which of the following order is incorrect :

(A) $H_2 < D_2 < T_2$ (Number of protons)

(B) $H_2 < D_2 < T_2$ (Bond energy)

(C) $H_2 < D_2 < T_2$ (Boiling point)

(D) $H_2 < D_2 < T_2$ (No. of neutrons)

Q3. Ortho and Para hydrogen differ :-

(A) In the number of protons

(B) In the molecular mass

(C) In the nature of spins of nucleus

(D) In the nature of spins of electrons

Q4. In which of the following reactions does hydrogen act as an oxidising agent :-

(A) $H_2 + F_2 \longrightarrow$

(B) $H_2 + SiCl_4 \longrightarrow$

(C) $Na + H_2 \longrightarrow$

(D) $CuO + H_2 \longrightarrow$

Q5. Temporary hardness may be removed from water by adding :-

(A) $CaCO_3$ (B) $Ca(OH)_2$

(C) $Ca(NO_3)_2$ (D) Hcl

Q6. Calgon is :-
(A) $Na_2[Na_4(PO_3)_6]$ (B) $Na_4[Na_2(PO_3)_6]$
(C) $Na_3[Na_3(PO_3)_3]$ (D) $Na[Na_2(PO_3)_4]$

Q7. Temporary hardness of water can be removed by :-
(A) addition of potassium permaganate
(B) boiling
(C) filtration
(D) addition of chlorine

Q8. A sample of water contains sodium chloride is :-
(A) Hard water (B) Softwater
(C) Moderately hard (D) None of these

Q9. Which substance can not be reduced by H_2O_2 :-
(A) $H^+/KMnO_4$ (B) $H^+/ K_2Cr_2O_4$
(C) Ag_2O (D) Fe^{+3}

Q10. Which of the following compounds is used for water softening?
(A) $Ca_3 (PO_4)_2$ (B) Na_3PO_4
(C) $Na_6P_6O_{18}$ (D) Na_2HPO_4

Q11. Calgon is an industrial name given to :-
(A) Normal sodium phosphate
(B) Sodium meta-aluminate
(C) Sodium hexametaphosphate
(D) Hydrated sodium aluminium silicate

Q12. H_2O_2, Molecular Geometry is :-
(A) Closed book structure
(B) Half open book structure
(C) full open book structure
(D) None

Q13. D_2O (heavy water) and H_2O differ in following except in :
(A) Freezing point
(B) Density
(C) Ionic product of water
(D) Its reaction with sodium

Q14. D_2O is preferred to H_2O, as a moderator, in nuclear reactors bcoz :-
(A) D_2O slows down fast neutrons better
(B) D_2O has high specific heat
(C) D_2O is cheaper
(D) None

Q15. Dihedral angle in H_2O_2 gas is :-
(A) 95° (B) 90°
(C) 112° (D) 101°

Q16. Decomposition of H_2O_2 can be prevented by addition of :
(A) Ferrous sulphate (B) oxalic acid
(C) Ozone (D) Glycerol

Q17. H_2O_2 acts as a reducing agent in its :-
(A) reaction with a ferrous salt
(B) reaction with iodides
(C) reaction with lead sulphide
(D) reaction with $KMnO_4$ in acid medium

Q18. Bond length of H–O in H_2O_2 solid is :-
(A) 0.99 A° (B) 1.45 A°
(C) 0.95 A° (D) 1.48 A°

Q19. Hydrogen peroxide is obtained by the electrolysis of _____.
(A) Water
(B) Sulphuric acid
(C) Hydrochloric acid
(D) Fused sodium peroxide

Q20. Hydrogen acts as a reducing agent and resembles with :-
(A) Halogen
(B) Noble gas
(C) Radioactive elements
(D) Alkali metals

Q21. In Bosch process which of the following gas uses for the production of H_2 gas
(A) Producer gas (B) Water gas
(C) Coal gas (D) Natural gas

Q22. Which of the following equations depict the oxidising nature of H_2O_2?

(A) $2MnO_4^- + 6H^+ \; 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_2$

(B) $2Fe^{3+} + 2H^+ + H_2O_2 \rightarrow 2Fe^{2+} + 2H_2O + O_3$

(C) $2I^- + 2H^+ + H_2O_2 \rightarrow I_2 + 2H_2O$

(D) $KIO_4 + H_2O_2 \rightarrow KIO_3 + H_2O + O_2$

Q23. An inorganic substance liberates oxygen on heating and turns an acidic solution of KI brown and reduces acidified $KMnO_4$ solution. The substance is :-

(A) HgO (B) H_2O_2

(C) KNO_3 (D) $Pb(NO_3)_2$

Exercise - 2A *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Dihydrogen can be prepared on commercial scale by different methods. In its preparation by the action of steam on hydrocarbons, a mixture of CO and H_2 gas is formed. It is known as _____.

(A) Water gas (B) Syngas

(C) Producer gas (D) Industrial gas

Q2. Which of the following statement(s) is/are correct in the case of heavy water?

(A) Heavy water is used as a moderator in nuclear reactor.

(B) Heavy water is more effective as solvent than ordinary water.

(C) Heavy water is more associated than ordinary water.

(D) Heavy water has lower boiling point than ordinary water.

PARAGRAPH TYPE

This section contains ***PARAGRAPHS****. Based on each paragraph, there are questions. Each question has FOUR options.* ***ONLY ONE*** *of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 3 to 5

A very small quantity of hydrogen peroxide is said to be formed when hydrogen and oxygen are exploded together.

The strength of hydrogen peroxide solution is stated in terms of the volume of oxygen evolved on heating.

$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

The commercial preparations are "10 to 20 volumes" according as it gives off 10 or 20 times its volume of oxygen. Merck's perhydrol contains about 30 percent of hydrogen peroxide corresponding with a "100 volume" strength. A "10 volume" solution would be roughly 3 percent strength.

Q3. There is a sample of 10 volume of hydrogen peroxide solution. Calculate its strength –

(A) 3.00% (B) 4.045%

(C) 2.509% (D) 3.035%

Q4. The volume strength of 1.5N H_2O_2 solution is :-

(A) 8.4 litres (B) 4.2 litres

(C) 16.8 litres (D) 5.2 litres

Q5. The strength in volumes of a solution containing 30.36 g/litres of H_2O_2 is –

(A) 10 volume

(B) 20 volume

(C) 5 volume

(D) None of these

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q6. Match List I with List II and select the correct answer using the codes given below the lists :-

List – I	**List II**
1. Heavy water	A. Bicarbonates of Mg and Ca in water
2. Temporary hardness of water	B. No foreign ions in water
3. Soft water	C. D_2O
4. Permanent hardness of water	D. Sulphates and water chlorides of Mg and Ca in water

(A) 1–C, 2 – D, 3 – B, 4 – A
(B) 1–B, 2 – A, 3 – C, 4 – D
(C) 1–B, 2 – D, 3 – C, 4 – A
(D) 1–C, 2 – A, 3 – B, 4 – D

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q7. Match the column :

Column I (For the production of H_2O_2)	**Column II**
(A) $BaO_2.8H_2O$	(p) Reduction and oxidation
(B) Na_2O_2	(q) Reacts with NaOH
(C) H_2SO_4	(r) Electrolysis then hydrolysis
(D) Ethyl derivative of anthraquinone	(s) Reacts with H_2SO_4

Q8. Match the column :

Column - I (Compound)	**Column - II (Properties)**
(A) $LiAlH_4$	(p) Binary hydride
(B) $NaBH_4$	(q) Reducing agent
(C) CaH_2	(r) Complex hydride
(D) LiH	(s) Alanate

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

Q9 **Statement 1 :** D_2O is called hard water.
Statement 2 : Its degree of dissociation is high.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
B	C	B	A	B	C	B	A	D	C
11	12	13	14	15	16	17	18	19	20
D	C	D	D	C	C	B	C	A	C
21	22	23							
B	B	D							

Ex-1A

1. C **2.** C, D **3.** A, D **4.** A, B **5.** A, B, C

6. C, D **7.** A **8.** A **9.** B

10. (A) → (q), (s) ; (B) → (r) ; (C) → (p), (r)

11. (A) → (s) ; (B) → (r) ; (C) → (q) ; (D) → (p)

12. (A) → (x), (y) ; (B) → (s), (t), (u), (v), (w) ; (C) → (t),(u) ;
(D) → (p), (r) ; (E) → (q), (s) ; (F) → (t)

13. A → s, B → r, C → p, D → q

14. 16 g **15.** D

Ex-2									
1	2	3	4	5	6	7	8	9	10
A	A	C	C	B	A	B	B	D	C
11	12	13	14	15	16	17	18	19	20
C	B	D	A	C	D	D	A	B	D
21	22	23							
B	C	B							

Ex-2A

1. A, B **2.** A, C **3.** D **4.** A **5.** A

6. D **7.** (A) → (iv) ; (B) → (iv) ; (C) → (iii) ; (D) → (i)

8. (A) → (ii), (iii), (iv) ; (B) → (ii), (iii) ; (C) → (i),(ii) ; (D) → (i), (ii)

9. D

-: Important Notes & Formulas :-